
Los Sabrossos

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1. Math

1.1. Sieve of Eratosthenes

```
1  const int N = 1e7;
2  int sieve[N];
3  void make(){
4      for(int i = 0; i < N; i++) sieve[i] = 1;
5      sieve[0] = sieve[1] = 0;
6      for (int i = 2; i < N; i++)
7          {
8              if(sieve[i]){
9                  for (int j = i*i; j < N; j+=i)
10                     {
11                         sieve[j] = 0;
12                     }
13             }
14         }
15 }
```

1.2. Euler Phi

```
1  ll phi(ll n) {
2      ll ans = n;
3      for (ll p = 2; p * p <= n; p++) {
4          if (n % p == 0) {
5              ans -= ans / p;
6              while (n % p == 0){
7                  n /= p;
8              }
9          }
10     }
11     if (n > 1){
12         ans -= ans / n;
13     }
14     return ans;
15 }
```

1.3. Binary Exponentiation

```
1  long long pote(long long a, long long b, long long mod){
2      long long ans = 1ll;
3      while(b){
4          if(b&1) ans = (ans * a)%mod;
5          a = (a * a)%mod;
6          b>>=1;
7      }
8      return ans;
9  }
10 // a^-1 = a^mod-2 = pote(a,mod-2);
```

1.4. Chinese Rest Theorem

```
1  //si los modulos son muy grandes 1e18, cambiar por int128 las operaciones
2  ll extgcd(ll a, ll b, ll &x, ll &y) {
3      if (!b) {
4          x = 1;
5          y = 0;
6          return a;
7      }
8
9      ll x1, y1;
10     ll g = extgcd(b, a % b, x1, y1);
11
12     x = y1;
13     y = x1 - (a / b) * y1;
14
15     return g;
16 }
17
18 // x      a (mod m)
19 // x      b (mod n)
20 //si retorna {0, -1} -> no solution
21 pair<ll,ll> CRT(ll a, ll m, ll b, ll n) {
22     ll x, y;
```

```

23     ll g = extgcd(m, n, x, y);
24
25     if ((b - a) % g != 0)
26         return {0, -1};
27
28     ll n2 = n / g;
29
30     ll t = ((b - a) / g * x) % n2;
31     if (t < 0) t += n2;
32
33     ll mod = m * n2;
34     ll r = (a + m * t) % mod;
35     if (r < 0) r += mod;
36
37     return {r, mod};
38 }
39
40 // x      a[i] (mod m[i]) para todo i
41 //vector de valores y vector de respectivos modulos
42 pair<ll,ll> CRT(vector<ll> a, vector<ll> m) {
43     ll r = a[0];
44     ll mod = m[0];
45
46     for (int i = 1; i < a.size(); i++) {
47         pair<ll,ll> res = CRT(r, mod, a[i], m[i]);
48
49         if (res.second == -1)
50             return {0, -1};
51
52         r = res.first;
53         mod = res.second;
54     }
55
56     return {r, mod};
57 }
58
59 void solve(){
60     vector<ll>a(2);
61     vector<ll>m(2);
62     repl(i,0,2){
63         cin >> a[i] >> m[i];
64     }
65     pair<ll,ll> ans = CRT(a, m);
66     if(ans.second == -1){
67         cout << "-1\n";
68     }
69     else{
70         cout << ans.first << " " << ans.second << "\n";
71     }
72 }

```

1.5. Extended Euclides

```

1 //Encuentra
2 //valores para x e y
3 //A*x + B*y = gcd(A, B)
4 ll extgcd(ll a, ll b, ll &x, ll &y){
5     if(b == 0){
6         x = 1;
7         y = 0;
8         return a;
9     }
10
11     ll x1, y1;
12     ll g = extgcd(b, a % b, x1, y1);
13     x = y1;
14     y = x1 - (a / b) * y1;
15     return g;
16 }
17
18 void solve(){
19     ll A, B, C;
20     cin >> A >> B >> C;
21     ll x, y;
22     ll g = extgcd(A, B, x, y);
23     if(C % g != 0){
24         cout << "-1\n";
25         return;
26     }
27     ll k = -C / g;
28     x *= k;
29     y *= k;
30     cout << x << " " << y << "\n";
31 }

```

1.6. Fibonacci Log(n)

```
1  long long int fibo(int n) {
2      long long int a, b, x, y, tmp;
3      a = x = tmp = 0;
4      b = y = 1;
5      while (n != 0)
6          if (n % 2 == 0) {
7              tmp = a * a + b * b;
8              b = b * a + a * b + b * b;
9              a = tmp;
10             n = n / 2;
11         } else {
12             tmp = a * x + b * y;
13             y = b * x + a * y + b * y;
14             x = tmp;
15             n = n - 1;
16         }
17         // x = fibo(n-1)
18         return y; // y = fibo(n)
19     }
```

1.7. Least Common Multiple

```
1  ll lcm(ll a, ll b){
2      return (b/__gcd(a,b)) * a;
3  }
```

1.8. Modular Inverse

```
1  const long long N = 1e6 + 10;
2  const int MOD = 1e9+7;
3  long long fact[N];
4  long long invfact[N];
5  void init(){
6      fact[0] = 1;
7      for(long long i = 1; i < N; i++)
8          fact[i] = fact[i-1] * i % MOD;
9
10     invfact[N-1] = pote(fact[N-1], MOD-2);
11
12     for(long long i = N-2; i >= 0; i--)
13         invfact[i] = invfact[i+1] * (i+1) % MOD;
14 }
15
16 long long nCk(long long n, long long k){
17     if(k < 0 || k > n) return 0;
18     return fact[n] * invfact[k] % MOD * invfact[n-k] % MOD;
19 }
```

1.9. Pollard Rho Rabin Miller

```
1  map<ll, ll> fact;
2
3  ll pote(ll a, ll b, ll mod) {
4      ll ans = 1;
5      while (b) {
6          if (b & 1) ans = (__int128)ans * a % mod;
7          a = (__int128)a * a % mod;
8          b >>= 1;
9      }
10     return ans;
11 }
12
13 ll mcd(ll a, ll b) {
14     a = abs(a);
15     b = abs(b);
16     while (a != 0) {
17         ll r = b % a;
18         b = a;
19         a = r;
20     }
21     return b;
22 }
```

```

22 }
23
24 //Rabin Miller
25 bool primo(ll n) {
26     if (n < 2) return false;
27     if (n == 2) return true;
28     ll D = n - 1, S = 0;
29     while (D % 2 == 0) {
30         D /= 2;
31         S++;
32     }
33     static const ll STEPS = 1611;
34     for(ll pasos = 0; pasos < STEPS; pasos++){
35         const ll A = 1 + rand() % (n - 1);
36         ll M = pote(A, D, n);
37         if (M == 1 || M == (n - 1)) goto next;
38         for(ll k = 0; k < S-1; k++){
39             M = (__int128)M * M % n;
40             if (M == (n - 1)) goto next;
41         }
42         return false;
43     next:;
44     }
45     return true;
46 }
47
48 //Rho de pollard
49 ll factor(ll n) {
50     static ll A, B;
51     A = 1 + rand() % (n - 1);
52     B = 1 + rand() % (n - 1);
53     #define fun(x) (((__int128)(x) * ((x) + B)) % n + A)
54
55     ll x = 2, y = 2, d = 1;
56     while (d == 1 || d == -1) {
57         x = fun(x);
58         y = fun(fun(y));
59         d = mcd(x - y, n);
60     }
61     return abs(d);
62 }
63
64 void factorize(ll n){
65     assert(n > 0);
66     while (n > 1 && !primo(n)) {
67         ll fx;
68         do {
69             fx = factor(n);
70         } while (fx == n);
71
72         n /= fx;
73         factorize(fx);
74
75         for (auto &it : fact) {
76             while (n % it.first == 0) {
77                 n /= it.first;
78                 it.second++;
79             }
80         }
81     }
82     if (n > 1) fact[n]++;
83 }

```

2. Geometry

2.1. Convex Hull

```

1 // Devuelve sentido horario
2 vector<point> hull(vector<point> p)
3 {
4     int n = p.size();
5     vector<point> h;
6     sort(p.begin(), p.end());
7     for(int i = 0; i < n; i++){
8         while(h.size() >= 2 and p[i].left(h[h.size() - 2], h.back()) < 0){
9             h.pop_back();
10        }
11        h.push_back(p[i]);
12    }
13    h.pop_back();
14    int k = h.size();

```

```

15     for(int i = n-1; i >= 0; i--)
16     {
17         while(h.size() >= k + 2 and p[i].left(h[h.size() - 2], h.back()) < 0){
18             h.pop_back();
19         }
20         h.pb(p[i]);
21     }
22     h.pop_back();
23     return h;
24 }

```

2.2. Point

```

1  struct point
2  {
3      int x, y;
4      point() {}
5      point(int x, int y): x(x), y(y) {}
6      point operator -(point p) {return point(x - p.x, y - p.y);}
7      point operator +(point p) {return point(x + p.x, y + p.y);}
8      int sq() const {return x * x + y * y;}
9      double abs() const {return sqrt(sq());}
10     int operator ^(point p) {return x * p.y - y * p.x;}
11     int operator *(point p) {return x * p.x + y * p.y;}
12     point operator *(int a) {return point(x * a, y * a);}
13     bool operator <(const point& p) const {return x == p.x ? y < p.y : x < p.x;}
14     int left(point a, point b) {
15         return ((b - a) ^ (*this - a)); // 0, -, +
16     }
17 };
18 ostream& operator<<(ostream& os, const point& p) {
19     return os << "(" << p.x << ", " << p.y << ")\n";
20 }
21
22
23 void polarSort(vector<point>& v) {
24     sort(v.begin(), v.end(), [] (point a, point b) {
25         const point origin{0, 0};
26         bool ba = a < origin, bb = b < origin;
27         if (ba != bb) { return ba < bb; }
28         return (a^b) > 0;
29     });
30 }

```

2.3. Closest Pair of Points Lineal

```

1  struct pt {
2      ll x, y;
3      bool operator == (const pt & o) const {
4          return x == o.x and y == o.y;
5      }
6  };
7
8  struct CustomHashPoint {
9      size_t operator()(const pt & p) const {
10         static
11         const uint64_t C =
12             chrono::steady_clock::now().time_since_epoch().count();
13         uint64_t x = p.x;
14         uint64_t y = p.y;
15         x ^= x >> 30;
16         x *= 0xbf58476d1ce4e5b9;
17         x ^= x >> 27;
18         x *= 0x94d049bb133111eb;
19         x ^= x >> 31;
20         y ^= y >> 30;
21         y *= 0xbf58476d1ce4e5b9;
22         y ^= y >> 27;
23         y *= 0x94d049bb133111eb;
24         y ^= y >> 31;
25         return C ^ (x + (y << 1));
26     }
27 };
28
29 ll dist2(pt a, pt b) {
30     ll x = a.x - b.x;
31     ll y = a.y - b.y;
32     return x * x + y * y;
33 }
34
35 pair < int, int > closest_pair(vector < pt > p) {

```



```

36     int n = p.size();
37     unordered_map < pt, int, CustomHashPoint > mp;
38     // puntos repetidos
39     for (int i = 0; i < n; i++) {
40         if (mp.count(p[i]))
41             return {
42                 mp[p[i]],
43                 i
44             };
45         mp[p[i]] = i;
46     }
47     mt19937 rng(
48         chrono::steady_clock::now().time_since_epoch().count()
49     );
50     auto rnd = [&]() {
51         return uniform_int_distribution < int > (0, n - 1)(rng);
52     };
53     ll d2 = dist2(p[0], p[1]);
54     pair < int, int > ans = {
55         0,
56         1
57     };
58     auto upd = [&](int i, int j) {
59         ll nd = dist2(p[i], p[j]);
60         if (nd < d2) {
61             d2 = nd;
62             ans = {
63                 i,
64                 j
65             };
66         }
67     };
68     // conseguir una distancia inicial razonable
69     for (int it = 0; it < 20; it++) {
70         int i = rnd();
71         int j = rnd();
72         while (i == j) j = rnd();
73         upd(i, j);
74     }
75     ll d = sqrtl((long double) d2) + 1;
76     unordered_map < pt, vector < int >, CustomHashPoint > grid;
77     grid.reserve(2 * n);
78     for (int i = 0; i < n; i++) {
79         grid[
80             p[i].x / d,
81             p[i].y / d
82         ].push_back(i);
83     }
84     for (auto & [c, v]: grid) {
85         // misma celda
86         for (int i = 0; i < (int) v.size(); i++) {
87             for (int j = i + 1; j < (int) v.size(); j++) {
88                 upd(v[i], v[j]);
89             }
90         }
91         // celdas vecinas
92         for (int dx = -1; dx <= 1; dx++) {
93             for (int dy = -1; dy <= 1; dy++) {
94                 if (dx == 0 and dy == 0) continue;
95                 pt to = {
96                     c.x + dx,
97                     c.y + dy
98                 };
99                 auto it = grid.find(to);
100                 if (it == grid.end()) continue;
101                 for (int i: v) {
102                     for (int j: it->second) {
103                         upd(i, j);
104                     }
105                 }
106             }
107         }
108     }
109     return ans;
110 }

```

2.4. Closest Pair of Points Nlogn

```

1 struct pt {
2     ll x, y;
3     int id;
4 };
5
6 ll dist2(pt a, pt b) {

```

```

7     ll x = a.x - b.x;
8     ll y = a.y - b.y;
9     return x * x + y * y;
10 }
11
12 pair < int, int > closest_pair(vector < pt > p) {
13     int n = p.size();
14     sort(p.begin(), p.end(), [](pt a, pt b) {
15         if (a.x != b.x) return a.x < b.x;
16         return a.y < b.y;
17     });
18     set < pair < ll, int >> s; // {y, id}
19     ll d2 = dist2(p[0], p[1]);
20     pair < int, int > ans = {
21         p[0].id,
22         p[1].id
23     };
24     int l = 0;
25     for (int i = 0; i < n; i++) {
26         ll d = sqrtl((long double) d2) + 1;
27         // sacar puntos que ya est n demasiado lejos en x
28         while (l < i and p[i].x - p[l].x >= d) {
29             s.erase({
30                 p[l].y,
31                 p[l].id
32             });
33             l++;
34         }
35         // buscar puntos con y cerca
36         auto it = s.lower_bound({
37             p[i].y - d,
38             -1
39         });
40         while (it != s.end() and it -> first <= p[i].y + d) {
41             int j = it -> second;
42             ll nd = dist2(p[i], p[j]);
43             if (nd < d2) {
44                 d2 = nd;
45                 ans = {
46                     p[i].id,
47                     p[j].id
48                 };
49                 d = sqrtl((long double) d2) + 1;
50             }
51             it++;
52         }
53         s.insert({
54             p[i].y,
55             p[i].id
56         });
57     }
58     return ans;
59 }

```

3. Strings

3.1. KMP

```

1 vector<int> kmp(string s){
2     int n = s.size();
3     vector<int> pi(n);
4     for(int i = 1; i < n; i++){
5         int j = pi[i-1];
6         while(j > 0 and s[i] != s[j]){
7             j = pi[j-1];
8         }
9         if(s[i] == s[j]){
10             j++;
11         }
12         pi[i] = j;
13     }
14     return pi;
15 }
16 //Pattern Matching
17 vector<int> ocurrencia(string texto, string patron) {
18     vector<int> P = kmp(patron);
19     int k = 0;
20     vector<int> M;
21     repl(i,0,texto.size()) {
22         while (k > 0 and patron[k] != texto[i]) k = P[k - 1];
23         if (patron[k] == texto[i]) k++;

```

```

24         if (k == patron.size()) {
25             M.pb(i - k + 1);
26             k = P[k - 1];
27         }
28     }
29     return M;
30 }
31
32 //Automata
33 void genAut(string &s) {
34     int n = s.size();
35
36     vector<vector<int>>> aut(n+1, vector<int>(26));
37     vector<int> pi = kmp(s);
38
39     for(int i = 0 ; i < n; i++){
40         aut[i][s[i]-'a'] = i+1;
41     }
42
43     for(int i = 0; i < n+1; i++){
44         for(int c = 0; c < 26; c++){
45             if (aut[i][c]) continue;
46             if (i < n and s[i] == 'a' + c) aut[i][c] = i+1;
47             else if (i > 0) aut[i][c] = aut[pi[i-1]][c];
48             else aut[i][c] = 0;
49         }
50     }
51 }

```

3.2. Z Function

```

1  vector<int> z_function(string s){
2      int n = s.size();
3      vector<int> z(n,0);
4      int l = 0;
5      int r = 0;
6      repl(i,l,n){
7          if(i < r){
8              z[i] = min(r - i, z[i - l]);
9          }
10         while(i + z[i] < n and s[z[i]] == s[i + z[i]]){
11             z[i]++;
12         }
13         if(i + z[i] > r){
14             l = i;
15             r = i + z[i];
16         }
17     }
18     return z;
19 }

```

3.3. Trie

```

1  const int E = 26;
2
3  struct AC {
4      int nodes;
5      vector<array<int, E>>go;
6      vector<bool>terminal;
7
8      AC(){
9          add_node();
10         nodes = 1;
11     }
12
13     void add_node(){
14         terminal.emplace_back();
15         go.emplace_back();
16     }
17
18     void insert(string &s){
19         int pos = 0;
20         for(char ch : s){
21             int c = ch - '0';
22             if(go[pos][c] == 0){
23                 go[pos][c] = nodes++;
24                 add_node();
25             }
26             pos = go[pos][c];
27         }
28         terminal[pos] = true;

```

```

29     }
30
31 };

```

3.4. Suffix Array

```

1  vector<int> suffix_array(string s) {
2      s += char(0);
3      const int n = s.size();
4      vector<int> a(n);
5      iota(a.begin(), a.end(), 0);
6      sort(a.begin(), a.end(), [&] (int i, int j){
7          return s[i] < s[j];
8      });
9      vector<int> mapping(n);
10     mapping[a[0]] = 0;
11     for(int i = 1; i < n; i++){
12         mapping[a[i]] = mapping[a[i-1]] + (s[a[i-1]] != s[a[i]]);
13     }
14     int len = 1;
15     vector<int> sbs(n);
16     vector<int> head(n);
17     vector<int> new_mapping(n);
18     while(len < n){
19         for(int i = 0; i < n; i++) sbs[i] = (a[i] + n - len) % n;
20         for(int i = n-1; i >= 0; i--) head[mapping[a[i]]] = i;
21         for(int i = 0; i < n; i++){
22             int x = sbs[i];
23             a[head[mapping[x]]++] = x;
24         }
25         new_mapping[a[0]] = 0;
26         for(int i = 1; i < n; i++){
27             if(mapping[a[i-1]] != mapping[a[i]]){
28                 new_mapping[a[i]] = new_mapping[a[i-1]] + 1;
29             }
30             else{
31                 int pre = mapping[(a[i-1] + len) % n];
32                 int cur = mapping[(a[i] + len) % n];
33                 new_mapping[a[i]] = new_mapping[a[i-1]] + (pre != cur);
34             }
35         }
36         len <<= 1;
37         swap(mapping, new_mapping);
38     }
39     return vector<int>(a.begin()+1, a.end()); //ignorar a[0] que es el centinela
40 }
41
42 vector<int> lcp_array(vector<int> &a, string s){
43     const int n = s.size();
44     vector<int> rank(n);
45     for(int i = 0; i < n; i++) rank[a[i]] = i;
46     int k = 0;
47     vector<int> lcp(n);
48     for(int i = 0; i < n; i++){
49         if(rank[i] == n-1){
50             k = 0;
51             continue;
52         }
53         int j = a[rank[i]+1];
54         while(i+k < n and j+k < n and s[i+k] == s[j+k]) k++;
55         lcp[rank[i]] = k;
56         if(k) k--;
57     }
58     lcp.pop_back();
59     return lcp;
60 }
61
62 void solve(){
63     //Numero de distintos substr en un string n*(n+1)/2 - sumatoria del LCP
64     //verificar si un string es substring de otro con BS en SA
65     //encontrar el longest common substring en 2 strings (ESTA IMPLEMENTACION)
66     string s, t;
67     cin >> s >> t;
68     if(s.size() > t.size()){
69         swap(s, t);
70     }
71     int n = s.size();
72     int m = t.size();
73     string st = s + '#' + t;
74     vector<int> sa = suffix_array(st);
75     vector<int> lcp = lcp_array(sa, st);
76     // print(sa);
77     // print(lcp);
78     vector<int> tr(n+m);

```

```

79     repl(i,0,n+m){
80         if(sa[i] >=n and sa[i+1]>=n){
81             tr[i] = 0;
82             continue;
83         }
84         if(sa[i] < n and sa[i+1] < n){
85             tr[i] = 0;
86             continue;
87         }
88         int mn = min(sa[i], sa[i+1]);
89         int tam1 = mn + lcp[i];
90         // dbg(tam1);
91         tr[i] = lcp[i];
92         if(tam1 > n){
93             dbg(tam1);
94             tr[i] = lcp[i] - (tam1-n);
95         }
96     }
97     // print(tr);
98     int mx = *max_element(tr.begin(), tr.end());
99     repl(i,0,n+m){
100         // dbg(tr[i]);
101         if(tr[i] == mx){
102             string ans = st.substr(sa[i], mx);
103             cout << ans << '\n';
104             return;
105         }
106     }
107 }
108 //Sabrosssus

```

3.5. Aho Corasick

```

1 //AhoCorasick para saber si un string s existe en un texto T
2 const int E = 26;
3 const int N = 1e6+10;
4 int cur;
5 int nodoTerminal[N]; //se guarda el nodo terminal del string i
6 set<int>st; //se guarda que nodos son terminales que aparecen en el texto T
7
8 struct AC {
9     int nodes;
10    vector<int>suf, super;
11    vector<array<int, E>>go;
12    vector<bool>terminal;
13    AC(){
14        add_node();
15        nodes = 1;
16    }
17    void add_node(){
18        terminal.emplace_back();
19        go.emplace_back();
20        super.emplace_back();
21        suf.emplace_back();
22    }
23
24    void insert(string &s){
25        int pos = 0;
26        for(char ch : s){
27            int c = ch - 'a';
28            if(go[pos][c] == 0){
29                go[pos][c] = nodes++;
30                add_node();
31            }
32            pos = go[pos][c];
33        }
34        terminal[pos] = true;
35        nodoTerminal[cur] = pos;
36    }
37
38    void build(){
39        queue<int>Q;
40        Q.emplace(0);
41        while(!Q.empty()){
42            int u = Q.front();
43            Q.pop();
44            super[u] = (suf[u] == 0 or terminal[suf[u]] ? suf[u] : super[suf[u]]);
45            for(int c = 0; c < E; c++){
46                if(go[u][c]){
47                    int v = go[u][c];
48                    Q.emplace(v);
49                    suf[v] = u == 0 ? 0 : go[suf[u]][c];
50                }

```

```

51         else{
52             go[u][c] = u == 0 ? 0 : go[suf[u]][c];
53         }
54     }
55 }
56 }
57 };
58
59 void mark(int u, AC &ac){
60     if(u == 0) return;
61     mark(ac.super[u], ac);
62     if(ac.terminal[u]){
63         st.insert(u);
64         ac.terminal[u] = false;
65     }
66     ac.super[u] = 0;
67 }
68
69 void solve(){
70     int n;
71     cin >> n;
72     AC ac;
73     repl(i,0,n){
74         string s;
75         cin >> s;
76         cur = i;
77         ac.insert(s);
78     }
79     string t; //texto grande
80     cin >> t;
81     ac.build();
82     int p = 0;
83     for(auto x : t){
84         int c = x - 'a';
85         p = ac.go[p][c];
86         mark(p, ac);
87     }
88     repl(i,0,n){
89         if(st.count(nodoTerminal[i])){
90             cout << "YES\n";
91         }
92         else{
93             cout << "NO\n";
94         }
95     }
96 }
97
98 //-----
99
100 //AhoCorasick para Frecuencias
101 const int E = 26;
102 const int N = 1e6+10;
103 int nodoTerminal[N];
104 int cur;
105 struct AC {
106     int nodes;
107     vector<int>suf, freq, by_level;
108     vector<bool>terminal;
109     vector<array<int,E>>go;
110
111     AC(){
112         add_node();
113         nodes = 1;
114     }
115     void add_node(){
116         terminal.emplace_back();
117         suf.emplace_back();
118         go.emplace_back();
119         freq.emplace_back();
120     }
121
122     void insert(string &s){
123         int pos = 0;
124         for(char ch:s){
125             int c = ch - 'a';
126             if(go[pos][c] == 0){
127                 go[pos][c] = nodes++;
128                 add_node();
129             }
130             pos = go[pos][c];
131         }
132         terminal[pos] = true;
133         nodoTerminal[cur] = pos;
134     }
135
136     void build(){

```

```

137     queue<int>Q;
138     Q.emplace(0);
139     while(!Q.empty()){
140         int u = Q.front();
141         Q.pop();
142         by_level.emplace_back(u);
143         for(int c = 0; c < E; c++){
144             if(go[u][c]){
145                 int v = go[u][c];
146                 Q.emplace(v);
147                 suf[v] = u == 0 ? 0 : go[suf[u]][c];
148             }
149             else{
150                 go[u][c] = u == 0 ? 0 : go[suf[u]][c];
151             }
152         }
153     }
154 }
155 };
156
157 void solve(){
158     int n;
159     cin >> n;
160     AC ac;
161     repl(i,0,n){
162         string s;
163         cin >> s;
164         cur = i;
165         ac.insert(s);
166     }
167     string t;
168     cin >> t;
169     ac.build();
170     int p = 0;
171     for(char ch:t){
172         int c = ch-'a';
173         p = ac.go[p][c];
174         ac.freq[p]++;
175     }
176     for(int i = (int)ac.by_level.size()-1; i > 0; i--){
177         int x = ac.by_level[i];
178         ac.freq[ac.suf[x]] += ac.freq[x];
179     }
180     repl(i,0,n){
181         int termi = nodoTerminal[i];
182         cout << ac.freq[termi] << '\n';
183     }
184 }

```

3.6. Booths Algorithm

```

1 //Algoritmo para menor rotacion lexicografica de un string en O(N)
2
3 int menorRotacion(string &s){
4     int n = s.size();
5     vector<int>f(2*n, -1);
6     int k = 0; //indice mejor rotacion
7     for(int j = 1; j < 2*n; j++){
8         int i = f[j-k-1];
9         //el operador %n simula la concatenacion de la cadena s+s
10        while(i!=-1 and s[j%n] != s[(k+i+1)%n]){
11            if(s[j%n] < s[(k+i+1)%n]){
12                k = j - i - 1;
13            }
14            i = f[i];
15        }
16        if(i == -1 and s[j%n] != s[(k+i+1)%n]){
17            if(s[j % n] < s[(k+i+1)%n]){
18                k = j;
19            }
20            f[j - k] = -1;
21        }
22        else{
23            f[j - k] = i + 1;
24        }
25    }
26    return k;
27 }
28 void solve(){
29     string s;
30     cin >> s;
31     ll n = s.size();
32     ll ans = menorRotacion(s);
33     repl(i, ans, ans+n){

```

```

34         cout << s[i%n];
35     }
36 }

```

3.7. Hashing 2D

```

1  /*
2   2D Hashing: buscar todas las apariciones de una matriz
3   r x c dentro de una matriz n x m.
4   */
5
6   const ll BASE = 911382323LL;
7
8   void solve() {
9       int r, c;
10      cin >> r >> c;
11
12      vector<string> pattern(r);
13      for (int i = 0; i < r; i++)
14          cin >> pattern[i];
15
16      // Potencias para el rolling vertical.
17      vector<ll> power(r + 1, 1);
18      for (int i = 1; i <= r; i++)
19          power[i] = power[i - 1] * BASE;
20
21      // Hash de la matriz patr n.
22      ll patternHash = 0;
23
24      for (int i = 0; i < r; i++) {
25          Hash h(pattern[i]);
26          ll rowHash = h.get(0, c);
27
28          patternHash += rowHash * power[r - i - 1];
29      }
30
31      int n, m;
32      cin >> n >> m;
33
34      vector<string> grid(n);
35      for (int i = 0; i < n; i++)
36          cin >> grid[i];
37
38      // rowHash[i][j] = hash de grid[i][j ... j+c-1].
39      vector<vector<ll>> rowHash(
40          n, vector<ll>(m - c + 1)
41      );
42
43      for (int i = 0; i < n; i++) {
44          Hash h(grid[i]);
45
46          for (int j = 0; j <= m - c; j++)
47              rowHash[i][j] = h.get(j, j + c);
48      }
49
50      // matrixHash[i][j] = hash de la submatriz r x c desde (i, j).
51      vector<vector<ll>> matrixHash(
52          n - r + 1,
53          vector<ll>(m - c + 1)
54      );
55
56      for (int j = 0; j <= m - c; j++) {
57          ll cur = 0;
58
59          // Primera ventana vertical.
60          for (int i = 0; i < r; i++)
61              cur += rowHash[i][j] * power[r - i - 1];
62
63          matrixHash[0][j] = cur;
64
65          // Rolling vertical.
66          for (int i = 1; i <= n - r; i++) {
67              cur -= rowHash[i - 1][j] * power[r - 1];
68              cur = cur * BASE + rowHash[i + r - 1][j];
69
70              matrixHash[i][j] = cur;
71          }
72      }
73
74      // 2D difference array para marcar las apariciones.
75      vector<vector<int>> diff(
76          n + 1, vector<int>(m + 1, 0)
77      );
78

```



```

79     for (int i = 0; i <= n - r; i++) {
80         for (int j = 0; j <= m - c; j++) {
81             if (matrixHash[i][j] == patternHash) {
82                 diff[i][j]++;
83                 diff[i][j + c]--;
84                 diff[i + r][j]--;
85                 diff[i + r][j + c]++;
86             }
87         }
88     }
89 }
90
91 // Prefijos 2D.
92 for (int i = 0; i < n; i++)
93     for (int j = 1; j < m; j++)
94         diff[i][j] += diff[i][j - 1];
95
96 for (int j = 0; j < m; j++)
97     for (int i = 1; i < n; i++)
98         diff[i][j] += diff[i - 1][j];
99
100 // Imprimir celdas pertenecientes a alguna aparici n.
101 for (int i = 0; i < n; i++) {
102     for (int j = 0; j < m; j++) {
103         cout << (diff[i][j] > 0 ? grid[i][j] : '.');
104     }
105     cout << '\n';
106 }
107 }

```

3.8. Hashing

```

1  struct Hash {
2      long long P=1777771,MOD[2],PI[2];
3      vector<long long> h[2],pi[2];
4
5      // SOLO PARA ROLLING / JOIN:
6      vector<long long> pw[2];
7
8      Hash(string& s){
9          MOD[0]=999727999;
10         MOD[1]=107077777;
11         PI[0]=325255434;
12         PI[1]=10018302;
13
14         for(int k = 0; k < 2; k++) {
15             h[k].resize(s.size()+1);
16             pi[k].resize(s.size()+1);
17
18             // SOLO PARA ROLLING / JOIN:
19             pw[k].resize(s.size()+1);
20         }
21
22         for(int k = 0; k < 2; k++){
23             h[k][0] = 0;
24             pi[k][0] = 1;
25
26             // SOLO PARA ROLLING / JOIN:
27             pw[k][0] = 1;
28
29             long long p = 1;
30
31             for(int i = 1; i < s.size()+1; i++){
32                 h[k][i] = (h[k][i-1]+p*s[i-1])%MOD[k];
33                 pi[k][i] = (1LL*pi[k][i-1]*PI[k])%MOD[k];
34
35                 // SOLO PARA ROLLING / JOIN:
36                 pw[k][i] = (1LL*pw[k][i-1]*P)%MOD[k];
37
38                 p = (p*P)%MOD[k];
39             }
40         }
41     }
42
43     long long get(int s, int e){
44         long long h0 = (h[0][e]-h[0][s]+MOD[0])%MOD[0];
45         h0 = (1LL*h0*pi[0][s])%MOD[0];
46
47         long long h1 = (h[1][e]-h[1][s]+MOD[1])%MOD[1];
48         h1 = (1LL*h1*pi[1][s])%MOD[1];
49
50         return (h0<<32)|h1;
51     }
52 }

```

```
53 // SOLO PARA ROLLING / JOIN:
54 long long join(long long a, long long b, int lenA){
55     long long a0 = a>>32;
56     long long a1 = a&0xffffffffLL;
57
58     long long b0 = b>>32;
59     long long b1 = b&0xffffffffLL;
60
61     a0 = (a0 + b0*pw[0][lenA])%MOD[0];
62     a1 = (a1 + b1*pw[1][lenA])%MOD[1];
63
64     return (a0<<32)|a1;
65 }
66 };
```

3.9. Manacher

```
1  const int N = 1e5 + 10;
2  int odd[N]; //odd[i] = max odd palindrome centered on i
3  int even[N]; //even[i] = max even palindrome centered on i
4  void manacher(string & s) {
5      int l = 0, r = -1, n = s.size();
6      for (int i = 0; i < n; i++) {
7          int k = i > r ? 1 : min(odd[l + r - i], r - i);
8          while (i + k < n and i - k >= 0 and s[i + k] == s[i - k]) k++;
9          odd[i] = k--;
10         if (i + k > r) l = i - k, r = i + k;
11     }
12     l = 0;
13     r = -1;
14     for (int i = 0; i < n; i++) {
15         int k = i > r ? 0 : min(even[l + r - i + 1], r - i + 1);
16         k++;
17         while (i + k <= n and i - k >= 0 and s[i + k - 1] == s[i - k]) k++;
18         even[i] = --k;
19         if (i + k - 1 > r) l = i - k, r = i + k - 1;
20     }
21 }
```

4. Bit Manipulation

4.1. Binary Representation

```
1  string tobit(long long x) {
2      string a(64, '0');
3      for (int i = 63; i >= 0; --i) {
4          if (x & 1) a[i] = '1';
5          x >>= 1;
6      }
7      return a;
8  }
```

4.2. Bit Mask

```
1  for(int mask = 0; mask < (1 << n); mask++) { //(1 << n) = 0(2^n)
2      for(int i = 0; i < n; i++) { // 0(n)
3          //verificamos si el i-esimo bit esta encendido
4          if(mask & (1 << i)) {
5              cout << v[i] << " ";
6          }
7      }
8  }
```

4.3. Bits Prendidos En Un Numero X En Una Posicion Pos

```
1  //retorna el numero de bits prendidos
2  //en una posicion pos desde 0 hasta el numero num
3  //tambien se puede usar en rangos asi:
4  //prendidos(R, 1) - prendidos(L-1, 1);
5  ll prendidos(ll num, ll pos){
6      if(num <= 0){
7          return 0;
```

```
8     }
9     ll pote = (1ll << pos);
10    ll blocks = (num + 1) / pote;
11    ll ones = blocks / 2;
12    ones *= pote;
13    if(blocks & 1){
14        ones += (num + 1) % pote;
15    }
16    return ones;
17 }
18 //Sabrosson
```

4.4. Turn Off a Bit at Position X

```
1 void solve(){
2     //mapa de caracteres en windows para ~
3     int a, x;
4     cin >> a >> x; // a es el numero, x es el numero de bit
5     //los bits se enumeran de dercha a izquierda iniciando de 0
6     /*
7     Ej: n = 8
8     Binario:      1 0 0 0
9     Posicion:     3 2 1 0
10    */
11    a = a & ~(1 << x); //apagamos el bit x de a
12    cout << a << "\n";
13 }
```

4.5. Xor Basis

```
1 ll base[LOG];
2 ll tam = 0;
3 void add(ll msk){
4     for(int i = LOG-1; i>= 0; i--){
5         if(!(msk&(1LL<<i)))continue;
6         if(!base[i]){
7             base[i]=msk;
8             tam++;
9             return;
10        }
11        msk^=base[i];
12    }
13 }
```

5. Data Structures

5.1. Segment Tree

```
1 const int N = 1e5;
2
3 struct info {
4     int num;
5 };
6
7 info ST[4 * N];
8
9 info ope(info u, info v) {
10     return {u.num + v.num};
11 }
12
13 void build(const vector<int> &v, int in = 1, int L = 0, int R = n - 1) {
14     if (L == R) {
15         ST[in] = {v[L]}; // Asignamos el valor desde el vector
16     } else {
17         int mid = (L + R) >> 1;
18         build(v, 2 * in, L, mid);
19         build(v, (2 * in) + 1, mid + 1, R);
20         ST[in] = ope(ST[2 * in], ST[(2 * in) + 1]);
21     }
22 }
23
24 void update(int pos, int val, int in = 1, int L = 0, int R = n - 1) {
25     if (L == R) {
26         ST[in].num = val;
27     } else {
```

```

28     int mid = (L + R) >> 1;
29     if (pos <= mid) {
30         update(pos, val, 2 * in, L, mid);
31     } else {
32         update(pos, val, (2 * in) + 1, mid + 1, R);
33     }
34     ST[in] = ope(ST[2 * in], ST[(2 * in) + 1]);
35 }
36 }
37
38 info get(int l, int r, int in = 1, int L = 0, int R = n - 1) {
39     if (r < L or R < l) return {0}; // Elemento neutro
40     if (l <= L and R <= r) return ST[in];
41
42     int mid = (L + R) >> 1;
43     info left = get(l, r, 2 * in, L, mid);
44     info right = get(l, r, (2 * in) + 1, mid + 1, R);
45     return ope(left, right);
46 }
47
48 // Llamadas simples y limpias:
49 // build(v); // Construye con el vector
50 // update(2, 10); // Cambia el valor en pos 2 a 10
51 // info ans = get(1, 3); // Obtiene la respuesta en el rango [1, 3]
52 // v = {1 2 3 4 5}
53 // get(1, 0, n-1, 1, 3) = 2 + 3 + 4 = 9

```

5.2. Segment Tree Iterativo

```

1  struct info
2  {
3      int num;
4  };
5
6  int sz;
7  vector<info> ST;
8
9  void build(int n) {
10     sz = 1;
11     while (sz < n) sz *= 2;
12     ST.assign(2 * sz, {1000000000});
13 }
14
15 void update(int pos, int val) {
16     pos += sz;
17     ST[pos].num = min(ST[pos].num, val);
18     for (pos /= 2; pos > 0; pos /= 2) {
19         ST[pos].num = min(ST[2 * pos].num, ST[2 * pos + 1].num);
20     }
21 }
22
23 info get(int l, int r) {
24     int res = 1000000000;
25     for (l += sz, r += sz + 1; l < r; l /= 2, r /= 2) {
26         if (l & 1) res = min(res, ST[l++].num);
27         if (r & 1) res = min(res, ST[--r].num);
28     }
29     return {res};
30 }
31 //Consultas exactamente igual que el recursivo

```

5.3. Segment Tree Lazy Propagation

```

1  const int N = 1e5;
2
3  struct info {
4      long long num;
5  };
6
7  info ST[4 * N];
8  long long lazy[4 * N];
9
10 info ope(info u, info v) {
11     return {u.num + v.num};
12 }
13
14 void build(const vector<int> &v, int in = 1, int L = 0, int R = n - 1) {
15     if (L == R) {
16         ST[in] = {v[L]};
17     } else {

```

```

18     int mid = (L + R) >> 1;
19
20     build(v, 2 * in, L, mid);
21     build(v, 2 * in + 1, mid + 1, R);
22
23     ST[in] = ope(ST[2 * in], ST[2 * in + 1]);
24 }
25 }
26
27 void push(int in, int L, int R) {
28     if (lazy[in] == 0)
29         return;
30
31     ST[in].num += lazy[in] * (R - L + 1);
32
33     if (L != R) {
34         lazy[2 * in] += lazy[in];
35         lazy[2 * in + 1] += lazy[in];
36     }
37
38     lazy[in] = 0;
39 }
40
41 void update(int l, int r, long long val,
42             int in = 1, int L = 0, int R = n - 1) {
43
44     push(in, L, R);
45
46     if (r < L or R < l)
47         return;
48
49     if (l <= L and R <= r) {
50         lazy[in] += val;
51         push(in, L, R);
52         return;
53     }
54
55     int mid = (L + R) >> 1;
56
57     update(l, r, val, 2 * in, L, mid);
58     update(l, r, val, 2 * in + 1, mid + 1, R);
59
60     ST[in] = ope(ST[2 * in], ST[2 * in + 1]);
61 }
62
63 info get(int l, int r,
64          int in = 1, int L = 0, int R = n - 1) {
65
66     push(in, L, R);
67
68     if (r < L or R < l)
69         return {0};
70
71     if (l <= L and R <= r)
72         return ST[in];
73
74     int mid = (L + R) >> 1;
75
76     info left = get(l, r, 2 * in, L, mid);
77     info right = get(l, r, 2 * in + 1, mid + 1, R);
78
79     return ope(left, right);
80 }

```

5.4. Bit 1D

```

1  struct BIT{
2      int n;
3      vector<ll> bit;
4      BIT(int _n){
5          n = _n;
6          bit.assign(n + 1, 0);
7      }
8      void add(int x, ll val){
9          for (int i = x; i <= n; i += (i & (-i))){
10             bit[i] += val;
11         }
12     }
13
14     ll sum(int x){
15         ll ans = 0;
16         for (int i = x; i > 0; i -= (i & (-i))){
17             ans += bit[i];
18         }
19     }
20 }

```

```

19     return ans;
20 }
21 ll query(int l, int r){
22     return sum(r) - sum(l - 1);
23 }
24 };
25 // Para actualizar en rango y consultas puntuales
26 // bit.add(l, val); bit.add(r+1, val);
27 // bit.query(l, pos)

```

5.5. Bit 2D

```

1 struct BIT{
2     int n, m;
3     vector<vector<ll>> bit;
4
5     BIT(int _n){
6         n = _n;
7         m = _n;
8         bit.assign(n+1, vector<ll>(n+1, 0));
9     }
10    void add(int x, int y, ll val){
11        for (int i = x; i <= n; i += (i&(-i))){
12            for (int j = y; j <= m; j += (j&(-j))){
13                bit[i][j] += val;
14            }
15        }
16    }
17    ll sum(int x, int y){
18        ll ans = 0ll;
19        for (int i = x; i > 0; i -= (i&(-i))){
20            for (int j = y; j > 0; j -= (j&(-j))){
21                ans += bit[i][j];
22            }
23        }
24        return ans;
25    }
26    ll query(int x1, int y1, int x2, int y2){
27        return sum(x2, y2) - sum(x2, y1 - 1) - sum(x1 - 1, y2) + sum(x1 - 1, y1 - 1);
28    }
29 };

```

5.6. Bit 2D Infinite Plane

```

1 struct BIT{
2     ll n;
3     vector<vector<ll>> xs;
4     vector<vector<ll>> bit;
5     BIT(ll nn){
6         n = nn;
7         xs.resize(n + 1);
8         bit.resize(n + 1);
9     }
10    void prepare(ll x, ll y){
11        for (ll i = x; i <= n; i += (i&(-i))){
12            xs[i].push_back(y);
13        }
14    }
15    void build(){
16        for (ll i = 1; i <= n; i++){
17            sort(xs[i].begin(), xs[i].end());
18            xs[i].erase(unique(xs[i].begin(), xs[i].end()), xs[i].end());
19            bit[i].assign(xs[i].size() + 1, 0);
20        }
21    }
22    void add(ll x, ll y, ll val){
23        for (ll i = x; i <= n; i += (i&(-i))){
24            ll p = lower_bound(xs[i].begin(), xs[i].end(), y) - xs[i].begin() + 1;
25            for (ll j = p; j < bit[i].size(); j += (j&(-j))){
26                bit[i][j] += val;
27            }
28        }
29    }
30    ll sum(ll x, ll y){
31        ll ans = 0ll;
32        for (ll i = x; i > 0; i -= (i&(-i))){
33            ll p = lower_bound(xs[i].begin(), xs[i].end(), y) - xs[i].begin();
34            for (ll j = p; j > 0; j -= (j&(-j))){
35                ans += bit[i][j];
36            }
37        }
38    }
39 };

```

```

38     return ans;
39 }
40 ll query(ll x1, ll y1, ll x2, ll y2){
41     return sum(x2-1,y2) - sum(x2-1, y1) - sum(x1 - 1, y2) + sum(x1 - 1, y1);
42 }
43
44 };
45 void solve(int tc){
46     int n,q; cin >> n >> q;
47     set<ll> X;
48     set<ll> Y;
49     vector<vector<ll>> points;
50     vector<vector<ll>> query;
51     repl(i,0,n){
52         ll x,y,w; cin >> x >> y >> w;
53         X.insert(x);
54         Y.insert(y);
55         points.push_back({x,y,w});
56     }
57     repl(i,0,q){
58         int t; cin >> t;
59         if(t == 0){
60             ll x,y,w; cin >> x >> y >> w;
61             X.insert(x);
62             Y.insert(y);
63             query.push_back({0,x,y,w});
64         }
65         else{
66             ll x1, y1, x2, y2;
67             cin >> x1 >> y1 >> x2 >> y2;
68             X.insert(x1);
69             X.insert(x2);
70             Y.insert(y1);
71             Y.insert(y2);
72             query.push_back({1,x1,y1,x2,y2});
73         }
74     }
75     map<ll,ll> equis, ye;
76     int i = 0;
77     for(ll x : X){
78         i++;
79         equis[x] = i;
80     }
81     i = 0;
82     for(ll x : Y){
83         i++;
84         ye[x] = i;
85     }
86     BIT bit((ll)X.size());
87     for(auto &x : points){
88         x[0] = equis[x[0]];
89         x[1] = ye[x[1]];
90         bit.prepare(x[0], x[1]);
91     }
92     for(auto &x : query){
93         if(x[0] == 0){
94             x[1] = equis[x[1]];
95             x[2] = ye[x[2]];
96             bit.prepare(x[1], x[2]);
97         }
98         else{
99             x[1] = equis[x[1]];
100             x[2] = ye[x[2]];
101             x[3] = equis[x[3]];
102             x[4] = ye[x[4]];
103         }
104     }
105     bit.build();
106     for(auto &x : points){
107         bit.add(x[0], x[1], x[2]);
108     }
109     for(auto &x : query){
110         if(x[0] == 0){
111             bit.add(x[1], x[2], x[3]);
112         }
113         else{
114             cout << bit.query(x[1],x[2],x[3],x[4]) << "\n";
115         }
116     }
117 }

```

5.7. Bit 3D

```

1 struct BIT {

```

```

2   int n;
3   int m;
4   int p;
5   vector < vector < vector < ll >>> bit;
6   BIT(int _n) {
7       n = _n + 10;
8       m = _n + 10;
9       p = _n + 10;
10      bit.assign(n, vector < vector < ll >> (n, vector < ll > (n, 0)));
11  }
12  void add(int x, int y, int z, int val) {
13      for (int i = x; i <= n; i += (i & (-i))) {
14          for (int j = y; j <= n; j += (j & (-j))) {
15              for (int k = z; k <= n; k += (k & (-k))) {
16                  bit[i][j][k] += val;
17              }
18          }
19      }
20  }
21  ll sum(int x, int y, int z) {
22      ll ans = 0 ll;
23      for (int i = x; i > 0; i -= (i & (-i))) {
24          for (int j = y; j > 0; j -= (j & (-j))) {
25              for (int k = z; k > 0; k -= (k & (-k))) {
26                  ans += bit[i][j][k];
27              }
28          }
29      }
30      return ans;
31  }
32  ll query(int x1, int y1, int z1, int x2, int y2, int z2) {
33      return sum(x2, y2, z2) - sum(x1 - 1, y2, z2) - sum(x2, y2, z1 - 1) - sum(x2, y1 - 1, z2)
34          + sum(x1 - 1, y1 - 1, z2) + sum(x2, y1 - 1, z1 - 1) + sum(x1 - 1, y2, z1 - 1) -
35          sum(x1 - 1, y1 - 1, z1 - 1);
36  }
37  };

```

5.8. DSU Dynamic Representative

```

1   int n, q;
2   cin >> n >> q;
3   DSU dsu(n + q + 10);
4   vector<int> id(n + 1), box(n + q + 10);
5   int nodes = n;
6   iota(id.begin(), id.end(), 0);
7   iota(box.begin(), box.end(), 0);
8   for(int i = 0 ; i < q ; i++){
9       int a, b;
10      cin >> a >> b;
11      int ori = id[a];
12      int ori2 = id[b];
13      dsu.parent[ori] = ori2;
14      nodes++;
15      id[a] = nodes;
16      box[nodes] = a;
17  }
18  for(int i = 1; i <= n ; i++){
19      cout << box[dsu.find(i)] << ' ';
20  }

```

5.9. Disjoint Set Union

```

1   struct DSU{
2       int num;
3       vector<int> parent;
4       vector<int> sizes;
5       DSU(){};
6       DSU(int n){
7           init(n);
8       }
9       void init(int n){
10          num = n;
11          parent.resize(n+1);
12          sizes.resize(n+1);
13          //si tienes index 0 inicializar desde 0
14          for(int i = 1; i <= n ; i++){
15              parent[i] = i;
16              sizes[i] = 1;
17          }
18      }
19      int find(int a){

```



```

20     if(a == parent[a]) return a;
21     return parent[a] = find(parent[a]);
22 }
23 void join(int a, int b){
24     int x = find(a);
25     int y = find(b);
26     if(x == y) return;
27     num--;
28     if(sizes[x] < sizes[y]){
29         parent[x] = y;
30         sizes[y] += sizes[x];
31     }
32     else{
33         parent[y] = x;
34         sizes[x] += sizes[y];
35     }
36 }
37 }
38 };
39 //Sabrossus

```

5.10. Dynamic Connectivity Offline

```

1  struct Change {
2      int a, b, sz_a;
3  };
4
5  struct DSU {
6      vector<int> parent, sz;
7      vector<Change> history;
8      int components;
9
10     DSU(int n) {
11         parent.resize(n + 1);
12         sz.assign(n + 1, 1);
13         iota(parent.begin(), parent.end(), 0);
14         components = n;
15     }
16
17     int find(int x) {
18         while (x != parent[x])
19             x = parent[x];
20         return x;
21     }
22
23     bool unite(int a, int b) {
24         a = find(a);
25         b = find(b);
26         if (a == b) {
27             history.push_back({-1, -1, -1});
28             return false;
29         }
30         if (sz[a] < sz[b])
31             swap(a, b);
32         history.push_back({a, b, sz[a]});
33         parent[b] = a;
34         sz[a] += sz[b];
35         components--;
36         return true;
37     }
38
39     int snapshot() {
40         return history.size();
41     }
42
43     void rollback(int snap) {
44         while (history.size() > snap) {
45             auto [a, b, old_sz_a] = history.back();
46             history.pop_back();
47             if (a == -1)
48                 continue;
49             parent[b] = b;
50             sz[a] = old_sz_a;
51             components++;
52         }
53     }
54 };
55
56 struct Edge {
57     int u, v;
58 };
59
60 struct SegmentTree {
61     int n;

```

```

62     vector<vector<Edge>> tree;
63     SegmentTree(int n) {
64         this->n = n;
65         tree.resize(4 * n + 5);
66     }
67     void add(int node, int l, int r, int ql, int qr, Edge e) {
68         if (qr < l or r < ql)
69             return;
70         if (ql <= l and r <= qr) {
71             tree[node].push_back(e);
72             return;
73         }
74         int mid = (l + r) / 2;
75         add(node * 2, l, mid, ql, qr, e);
76         add(node * 2 + 1, mid + 1, r, ql, qr, e);
77     }
78     void add(int l, int r, Edge e) {
79         if (l > r)
80             return;
81         add(1, 1, n, l, r, e);
82     }
83 };
84
85 void dfs(int node, int l, int r, SegmentTree &seg, DSU &dsu, vector<int> &answer){
86     int snap = dsu.snapshot();
87     for (auto [u, v] : seg.tree[node])
88         dsu.unite(u, v);
89     if (l == r) {
90         answer[l] = dsu.components;
91     } else {
92         int mid = (l + r) / 2;
93         dfs(node * 2, l, mid, seg, dsu, answer);
94         dfs(node * 2 + 1, mid + 1, r, seg, dsu, answer);
95     }
96     dsu.rollback(snap);
97 }
98
99 void solve() {
100     int n, q;
101     cin >> n >> q;
102     if (q == 0)
103         return;
104     DSU dsu(n);
105     SegmentTree seg(q);
106     map<pair<int,int>, int> start;
107     vector<int> answer(q + 1);
108     vector<bool> query(q + 1);
109     for (int t = 1; t <= q; t++) {
110         char type;
111         cin >> type;
112         if (type == '?') {
113             query[t] = true;
114         } else {
115             int u, v;
116             cin >> u >> v;
117             if (u > v)
118                 swap(u, v);
119             if (type == '+') {
120                 start[{u, v}] = t;
121             } else {
122                 int L = start[{u, v}];
123                 seg.add(L, t - 1, {u, v});
124                 start.erase({u, v});
125             }
126         }
127     }
128     for (auto [edge, L] : start) {
129         auto [u, v] = edge;
130         seg.add(L, q, {u, v});
131     }
132     dfs(1, 1, q, seg, dsu, answer);
133     for (int t = 1; t <= q; t++) {
134         if (query[t])
135             cout << answer[t] << '\n';
136     }
137 }

```

5.11. Implicit Treap

```

1 // Mantiene las posiciones
2 struct Node {
3     int val , pri , sz ;
4     Node *l , *r;
5     Node ( int v) {

```

```

6         val = v;
7         pri = rand () ;
8         sz = 1;
9         l = r = nullptr ;
10    }
11 };
12 using pnode = Node *;
13 int sz ( pnode t) {
14     return t ? t -> sz : 0;
15 }
16 void upd (pnode t) {
17     if (t) t -> sz = 1 + sz (t -> l) + sz (t -> r);
18 }
19 void split ( pnode t , pnode &l , pnode &r , int k) {
20     if (! t) {
21         l = r = nullptr ;
22         return ;
23     }
24     int leftSize = sz (t -> l);
25     if ( leftSize >= k) {
26         split (t -> l , l , t -> l , k) ;
27         r = t;
28     }
29     else {
30         split (t -> r , t -> r , r , k - leftSize - 1) ;
31         l = t;
32     }
33     upd (t);
34 }
35 pnode merge ( pnode l , pnode r) {
36     if (! l || ! r ) return l ? l : r;
37
38     if (l -> pri > r -> pri ) {
39         l -> r = merge (l -> r , r);
40         upd (l );
41         return l;
42     }
43     else {
44         r -> l = merge (l , r -> l );
45         upd (r );
46         return r;
47     }
48 }
49 void insert ( pnode & root , int pos , int val ) {
50     pnode a , b;
51     split ( root , a , b , pos );
52     root = merge ( merge (a , new Node ( val )) , b) ;
53 }
54 void erase ( pnode & root , int pos ) {
55     pnode a , b , c;
56     split ( root , a , b , pos );
57     split (b , b , c , 1) ;
58     root = merge (a , c) ;
59 }
60 void print(pnode t) {
61     if (!t) return;
62
63     print(t->l);
64     cout << t->val << " ";
65     print(t->r);
66 }
67 int get(pnode t, int k) {
68     if (!t) return -1;
69
70     int leftSize = sz(t->l);
71
72     if (k < leftSize)
73         return get(t->l, k);
74
75     if (k == leftSize)
76         return t->val;
77
78     return get(t->r, k - leftSize - 1);
79 }

```

5.12. Ordered Set

```

1  #include <bits/stdc++.h>
2  #include <ext/pb_ds/assoc_container.hpp>
3  #include <ext/pb_ds/tree_policy.hpp>
4
5  using namespace std;
6  using namespace __gnu_pbds;
7

```

```

8  typedef tree<int, null_type, less<int>, rb_tree_tag, tree_order_statistics_node_update>
   ordered_set;
9
10 void solve() {
11     ordered_set st;
12
13     st.insert(10);
14     st.insert(30);
15     st.insert(20);
16
17     // 1. Elemento en la posición K (0-indexed, de menor a mayor)
18     // find_by_order devuelve un iterador (puntero). Usamos '*' para el valor.
19     auto it = st.find_by_order(1);
20     if (it != st.end()) {
21         int elemento = *it; // Devuelve 20 de forma segura
22     }
23
24     // 2. Cuantos elementos son estrictamente menores que X
25     int menores_que_25 = st.order_of_key(25); // Devuelve 2 (el 10 y el 20)
26 }
27 // Sabrossus

```

5.13. Range Bit

```

1  struct RangeBIT{
2      int n;
3      vector<ll> bit1, bit2;
4      RangeBIT(int nn){
5          n = nn;
6          bit1.assign(n + 1, 0);
7          bit2.assign(n + 1, 0);
8      }
9      void add_one(vector<ll> &b, int x, ll val){
10         for (int i = x; i <= n; i += (i & (-i))) {
11             b[i] += val;
12         }
13     }
14     void add(int l, int r, ll val){
15         add_one(bit1, l, val);
16         add_one(bit1, r + 1, -val);
17         add_one(bit2, l, l * val);
18         add_one(bit2, r + 1, -(r + 1) * val);
19     }
20     ll sum_one(const vector<ll> &b, int x){
21         ll ans = 0;
22         for (int i = x; i > 0; i -= (i & (-i))) {
23             ans += b[i];
24         }
25         return ans;
26     }
27
28     ll sum(int x){
29         return (x + 1) * sum_one(bit1, x) - sum_one(bit2, x);
30     }
31     ll query(int l, int r){
32         return sum(r) - sum(l - 1);
33     }
34 };

```

5.14. Sparse Table

```

1  const int N = 1e5;
2  const int LOG = 20;
3  int v[N]; // values
4  int st[N][LOG]; // Sparse Table
5  int lg[N]; // log values
6  int n;
7  void build(){
8      for(int i = 0; i < n; i++) st[i][0] = v[i];
9
10     for(int i = 1; i < LOG; i++){
11         for(int j = 0; j < n - (1<<i) + 1; j++){
12             st[j][i] = min(st[j][i-1], st[j+(1<<(i-1))][i-1]);
13         }
14     }
15     lg[1] = 0;
16     for(int i = 2; i <= n; i++) lg[i] = lg[i/2] + 1;
17 }
18 int query(int L, int R){
19     int k = lg[R-L+1];
20     return min(st[L][k], st[R-(1<<k)+1][k]);

```

```

21 }
22 // v = [5, 2, 4, 7, 1, 3]
23 // query(1, 4)      [2, 4, 7, 1]

```

5.15. Treap

```

1 // Ordena los elementos por valor
2 struct Node {
3     int key, pri, sz;
4     Node *l, *r;
5     Node(int k) {
6         key = k;
7         pri = rand();
8         sz = 1;
9         l = r = nullptr;
10    }
11 };
12
13 using pnode = Node*;
14
15 int sz(pnode t) {
16     return t ? t->sz : 0;
17 }
18
19 void upd(pnode t) {
20     if (t) t->sz = 1 + sz(t->l) + sz(t->r);
21 }
22
23 void split(pnode t, pnode &l, pnode &r, int k) {
24     if (!t) {
25         l = r = nullptr;
26         return;
27     }
28     int leftSize = sz(t->l);
29     if (leftSize >= k) {
30         split(t->l, l, t->l, k);
31         r = t;
32     } else {
33         split(t->r, t->r, r, k - leftSize - 1);
34         l = t;
35     }
36     upd(t);
37 }
38
39 pnode merge(pnode l, pnode r) {
40     if (!l || !r) return l ? l : r;
41
42     if (l->pri > r->pri) {
43         l->r = merge(l->r, r);
44         upd(l);
45         return l;
46     } else {
47         r->l = merge(l, r->l);
48         upd(r);
49         return r;
50     }
51 }
52
53 void insert(pnode &root, int pos, int val) {
54     pnode a, b;
55     split(root, a, b, pos);
56     root = merge(merge(a, new Node(val)), b);
57 }
58
59 void erase(pnode &root, int pos) {
60     pnode a, b, c;
61     split(root, a, b, pos);
62     split(b, b, c, 1);
63     root = merge(a, c);
64 }
65
66 int get(pnode t, int k) {
67     if (!t) return -1;
68
69     int leftSize = sz(t->l);
70
71     if (k < leftSize)
72         return get(t->l, k);
73
74     if (k == leftSize)
75         return t->key;
76 }

```

```
79     return get(t->r, k - leftSize - 1);
80 }
81
82 void print(pnode t) {
83     if (!t) return;
84
85     print(t->l);
86     cout << t->key << " ";
87     print(t->r);
88 }
89 }
```

6. Dynamic Programing

6.1. Bounded Knapsack

```
1  // bounded_knapsack para DPs del estilo :
2  // dp[j - cw] + cv , w-> pesos y quieres maximizar el valor
3  int n, we;
4  cin >> n >> we;
5  vector<int> v, m, w, dp;
6  v.resize(n);
7  m.resize(n);
8  w.resize(n);
9  dp.assign(we+1, 0);
10 int sum = 0;
11 for(int i = 0 ; i < n ; i++){
12     int a, b, c;
13     cin >> a >> b >> c;
14     v[i] = a;
15     w[i] = b;
16     m[i] = c;
17     vector<int> old = dp;
18     for(int r = 0; r < w[i] and r <= we ; r++){
19         deque<pair<int,int>> dq;
20         for(int j = r; j <= we ; j += w[i]){
21             int k = (j - r)/w[i];
22             while(!dq.empty() and dq.front().first < k - m[i]){
23                 dq.pop_front();
24             }
25             int cur = old[j] - k * v[i];
26             while(!dq.empty() && dq.back().second <= cur){
27                 dq.pop_back();
28             }
29             dq.push_back({k, cur});
30             dp[j] = dq.front().second + k * v[i];
31         }
32     }
33 }
34 cout << dp[we];
```

6.2. Kadane 2D

```
1  int n;
2  cin >> n;
3  vector < vector < int >> v(n);
4  repl(i, 0, n) {
5      repl(j, 0, n) {
6          int x;
7          cin >> x;
8          v[i].push_back(x);
9      }
10 }
11 int maxi = INT_MIN;
12 repl(ini, 0, n) {
13     vector < int > pref(n, 0);
14     repl(i, ini, n) {
15         repl(j, 0, n) {
16             pref[j] += v[i][j];
17         }
18         int cur = 0;
19         repl(j, 0, n) {
20             cur = max(pref[j], pref[j] + cur);
21             maxi = max(maxi, cur);
22         }
23     }
24 }
```

6.3. Kadane 3D

```

1  int n;
2  cin >> n;
3
4  repl(i, 0, n) {
5      repl(j, 0, n) {
6          repl(k, 0, n) {
7              cin >> v[i][j][k];
8          }
9      }
10 }
11 int maxi = INT_MIN;
12 repl(k_ini, 0, n) {
13     vector < vector < int >> pref2(n, vector < int > (n, 0));
14     repl(k, k_ini, n) {
15         repl(i, 0, n) {
16             repl(j, 0, n) {
17                 pref2[i][j] += v[i][j][k];
18             }
19         }
20         repl(ini, 0, n) {
21             vector < int > pref(n, 0);
22             repl(i, ini, n) {
23                 repl(j, 0, n) {
24                     pref[j] += pref2[i][j];
25                 }
26                 int cur = 0;
27                 repl(u, 0, n) {
28                     cur = max(pref[u], cur + pref[u]);
29                     maxi = max(maxi, cur);
30                 }
31             }
32         }
33     }
34 }
35 }
36
37 cout << maxi << "\n";

```

6.4. SOS DP

```

1  void sos_dp(vector<ll>& dp, int K){
2      // k es el tamaño del arreglo :v
3      for(int bit = 0; bit < K; bit++){
4          for(int mask = 0; mask < (1 << K); mask++){
5              if(mask & (1 << bit)){
6                  dp[mask] += dp[mask ^ (1 << bit)];
7                  // si quieres que sea un sos dp de máximos
8                  // dp[mask] = max(dp[mask], dp[mask ^ (1 << bit)]);
9              }
10         }
11     }
12 }

```

6.5. Fastfib

```

1  #include <bits/stdc++.h>
2  #define ll long long
3
4  using namespace std;
5
6  const ll MOD = 1e9 + 7;
7
8  struct Matrix {
9      ll a[2][2];
10     Matrix() {
11         memset(a, 0, sizeof(a));
12     }
13 };
14
15 Matrix multiply(Matrix A, Matrix B) {
16     Matrix C;
17     for (int i = 0; i < 2; i++) {
18         for (int j = 0; j < 2; j++) {
19             for (int k = 0; k < 2; k++) {
20                 C.a[i][j] = (C.a[i][j] + A.a[i][k] * B.a[k][j]) % MOD;
21             }
22         }
23     }
24 }

```

```

24     return C;
25 }
26
27 Matrix power(Matrix A, ll n) {
28     Matrix R;
29     R.a[0][0] = 1;
30     R.a[1][1] = 1;
31     while (n) {
32         if (n & 1)
33             R = multiply(R, A);
34         A = multiply(A, A);
35         n >>= 1;
36     }
37     return R;
38 }
39
40 ll fibonacci(ll n) {
41     if (n == 0) return 0;
42     Matrix A;
43     A.a[0][0] = 1;
44     A.a[0][1] = 1;
45     A.a[1][0] = 1;
46     A.a[1][1] = 0;
47     Matrix R = power(A, n - 1);
48     return R.a[0][0];
49 }
50
51 int main() {
52     ll n;
53     cin >> n;
54     cout << fibonacci(n) << '\n';
55 }

```

7. Flow

7.1. Dinic

```

1  struct Dinic
2  {
3      const int INF = 1e9;
4
5      struct flowEdge
6      {
7          int to, rev, f, cap;
8      };
9
10     vector<vector<flowEdge>> G;
11     vector<int> work, lvl, Q;
12
13     int S, T, nodes;
14
15     Dinic(){}
16     Dinic(int n, int s, int t){
17         S = s;
18         T = t;
19         nodes = n;
20         G.clear();
21         G.resize(n);
22         Q.resize(n);
23     }
24     void addEdge(int st, int en, int cap){
25         flowEdge A = {en, (int)G[en].size(), 0, cap};
26         flowEdge B = {st, (int)G[st].size(), 0, 0};
27         G[st].push_back(A);
28         G[en].push_back(B);
29     }
30     int dfs(int v, int f){
31         if(v == T or f == 0) return f;
32         for(int &i = work[v]; i < G[v].size(); i++){
33             flowEdge &e = G[v][i];
34             int u = e.to;
35             if(e.cap <= e.f or lvl[u] != lvl[v] + 1) continue;
36             int df = dfs(u, min(f, e.cap - e.f));
37             if(df){
38                 e.f += df;
39                 G[u][e.rev].f -= df;
40                 return df;
41             }
42         }
43         return 0;
44     }

```



```

45     bool bfs(){
46         int qt = 0;
47         Q[qt++] = S;
48         lvl.assign(nodes, -1);
49         lvl[S] = 0;
50         for(int i = 0; i < qt; i++){
51             int u = Q[i];
52             for(flowEdge &e : G[u]){
53                 int v = e.to;
54                 if(e.cap <= e.f or lvl[v] != -1) continue;
55                 lvl[v] = lvl[u] + 1;
56                 Q[qt++] = v;
57             }
58         }
59         return lvl[T] != -1;
60     }
61     int maxFlow(){
62         int flow = 0;
63         while(bfs()){
64             work.assign(nodes, 0);
65             while(true){
66                 int df = dfs(S, INF);
67                 if(df == 0) break;
68                 flow += df;
69             }
70         }
71         return flow;
72     }
73 }
74 };

```

8. Graph Algorithms

8.1. Dijkstra

```

1  const int N = 1e5;
2
3  int vis[N];
4  vector<pair<int,int>> G[N];
5
6  void dijkstra(int x){
7      priority_queue<pair<int,int>, vector<pair<int,int>>, greater<pair<int,int>>> pq;
8      pq.push({0,x});
9      vis[x] = 0;
10     while(!pq.empty()){
11         auto y = pq.top();
12         pq.pop();
13         int u = y.second;
14         int cost = y.first;
15         for(auto v : G[u]){
16             int cur = cost + v.second;
17             if(cur < vis[v.first]){
18                 vis[v.first] = cur;
19                 pq.push({cur,v.first});
20             }
21         }
22     }
23 }

```

8.2. Kruskal MST

```

1  const int N = 1e5;
2  struct DSU{
3      int num;
4      vector<int> parent;
5      vector<int> sizes;
6      DSU(){};
7      DSU(int n){
8          init(n);
9      }
10     void init(int n){
11         num = n;
12         parent.resize(n+1);
13         sizes.resize(n+1);
14         for(int i = 1; i <= n ; i++){
15             parent[i] = i;
16             sizes[i] = 1;
17         }

```

```

18     }
19     int find(int a){
20         if(a == parent[a]) return a;
21         return parent[a] = find(parent[a]);
22     }
23     void join(int a, int b){
24         int x = find(a);
25         int y = find(b);
26         if(x == y) return;
27         num--;
28         if(sizes[x] < sizes[y]){
29             parent[x] = y;
30             sizes[y] += sizes[x];
31         }
32         else{
33             parent[y] = x;
34             sizes[x] += sizes[y];
35         }
36     }
37 };
38 int n;
39 vector<pair<int, pair<int, int>>> List;
40 int Kruskal_MST(){
41     sort(List.begin(), List.end());
42     DSU dsu(n);
43     int mst = 0, t = 1;
44     for(int i = 0; i < List.size(); i++){
45         int w = List[i].first; //weight
46         int a = List[i].second.first; //at
47         int b = List[i].second.second; //to
48         if(dsu.find(a) != dsu.find(b)){
49             dsu.join(a, b);
50             mst += w;
51             t++;
52         }
53         if(t == n) break;
54     }
55     return mst;
56 }

```

8.3. Topological Sort Kahn Algorithm

```

1  const int N = 1e5;
2
3  int indegree[N];
4  vector<int> G[N];
5  int n;
6  vector<int> Kahn(){
7      queue<int> q;
8      vector<int> ans;
9      for(int i = 0; i < n; i++){
10         if(indegree[i] == 0) q.push(i);
11     }
12     while (!q.empty()){
13         int v = q.front();
14         q.pop();
15         ans.push_back(v);
16         for(int u : G[v]){
17             indegree[u]--;
18             if(indegree[u] == 0){
19                 q.push(u);
20             }
21         }
22     }
23     //if(ans.size() != n) grafo no DAG
24     return ans;
25 }

```

8.4. Kosaraju

```

1  struct Kosaraju {
2      int n, components;
3      vector<vector<int>> graph, invgraph;
4      vector<int> vis, order, component;
5
6      Kosaraju(int n) : n(n) {
7          graph.resize(n + 1);
8          invgraph.resize(n + 1);
9          vis.resize(n + 1);
10         component.resize(n + 1);
11     }

```

```

12
13     void add_edge(int u, int v) {
14         graph[u].push_back(v);
15         invgraph[v].push_back(u);
16     }
17
18     void dfs1(int u) {
19         vis[u] = 1;
20         for (int v : graph[u]) {
21             if (!vis[v])
22                 dfs1(v);
23         }
24         order.push_back(u);
25     }
26
27     void dfs2(int u, int id) {
28         vis[u] = 1;
29         component[u] = id;
30         for (int v : invgraph[u]) {
31             if (!vis[v])
32                 dfs2(v, id);
33         }
34     }
35     void build() {
36         for (int u = 1; u <= n; u++) {
37             if (!vis[u])
38                 dfs1(u);
39         }
40         reverse(order.begin(), order.end());
41         fill(vis.begin(), vis.end(), 0);
42         components = 0;
43         for (int u : order) {
44             if (!vis[u]) {
45                 components++;
46                 dfs2(u, components);
47             }
48         }
49     }
50 };

```

8.5. Kuhn

```

1  int n;
2  vector<vector<int>> G;
3  vector<int> mt;
4  vector<bool> used;
5  bool try_kuhn(int v){
6      if(used[v]) return false;
7      used[v] = true;
8      for(int to : G[v]){
9          if(mt[to] == -1 or try_kuhn(mt[to])){
10             mt[to] = v;
11             return true;
12         }
13     }
14     return false;
15 }
16 void solve(int tc){
17     mt.assign(n, -1);
18     repl(v,0,n){
19         used.assign(n, false);
20         try_kuhn(v);
21     }
22 }

```

8.6. 2SAT

```

1  struct TwoSAT {
2      int n, nodes;
3      vector<vector<int>> graph, invgraph;
4      vector<int> vis, order, component;
5      vector<bool> ans;
6
7      TwoSAT(int n) : n(n), nodes(2 * n) {
8          graph.resize(nodes);
9          invgraph.resize(nodes);
10         vis.resize(nodes);
11         component.resize(nodes);
12         ans.resize(n);
13     }
14 }

```

```

15     int id(int x, bool value) {
16         return 2 * x + value;
17     }
18
19     int neg(int x) {
20         return x ^ 1;
21     }
22
23     void add_implication(int a, int b) {
24         graph[a].push_back(b);
25         invgraph[b].push_back(a);
26     }
27
28     void add_or(int a, int b) {
29         add_implication(neg(a), b);
30         add_implication(neg(b), a);
31     }
32
33     void add_xor(int a, int b) {
34         add_or(a, b);
35         add_or(neg(a), neg(b));
36     }
37
38     void add_equivalence(int a, int b) {
39         add_implication(a, b);
40         add_implication(b, a);
41         add_implication(neg(a), neg(b));
42         add_implication(neg(b), neg(a));
43     }
44
45     void dfs1(int u) {
46         vis[u] = 1;
47         for (int v : graph[u]) {
48             if (!vis[v])
49                 dfs1(v);
50         }
51         order.push_back(u);
52     }
53
54     void dfs2(int u, int id) {
55         vis[u] = 1;
56         component[u] = id;
57         for (int v : invgraph[u]) {
58             if (!vis[v])
59                 dfs2(v, id);
60         }
61     }
62
63     bool solve() {
64         for (int u = 0; u < nodes; u++) {
65             if (!vis[u])
66                 dfs1(u);
67         }
68         reverse(order.begin(), order.end());
69         fill(vis.begin(), vis.end(), 0);
70         int cnt = 0;
71         for (int u : order) {
72             if (!vis[u]) {
73                 dfs2(u, cnt++);
74             }
75         }
76         for (int i = 0; i < n; i++) {
77             if (component[2 * i] == component[2 * i + 1])
78                 return false;
79             ans[i] = component[2 * i] > component[2 * i + 1];
80         }
81         return true;
82     }
83 };

```

9. Tree Algorithms

9.1. Euler Tour

```

1     const int N = 1e5 + 10;
2     vector<int> path;
3     vector<int> G[N];
4     void EulerTour(int node, int parent){
5         path.pb(node);
6         for(int u : G[node]){
7             if(u != parent){

```

```

8         EulerTour(u,node);
9     }
10 }
11 path.pb(node);
12 }
13
14 const int N = 1e5 + 10;
15 int tin[N];
16 int tout[N];
17 int flat[N];
18 int timer = 1;
19 void EulerTour(int node, int parent){
20     tin[node] = timer;
21     flat[timer] = node;
22     timer++;
23     for(int u : G[node]){
24         if(u != parent){
25             EulerTour(u, node);
26         }
27     }
28     tout[node] = timer;
29 }

```

9.2. Binary Lifting

```

1  const int N = 1e5;
2  const int LOG = 20;
3  int up[N][LOG];
4  vector<int> G[N];
5  //Find parents
6  void dfs(int u){
7      for(int v : G[u]){
8          up[v][0] = u;
9          for(int j = 1; j < LOG; j++){
10             up[v][j] = up[up[v][j-1]][j-1];
11         }
12         dfs(v);
13     }
14 }
15 int get_K_parent(int a, int k){
16     for(int j = LOG-1; j >= 0; j++){
17         if(k & (1<<j)){
18             a = up[a][j];
19         }
20     }
21     return a;
22 }

```

9.3. Centroid Decomposition

```

1  //EJERCICIO IMPLEMENTACION XENIA AND TREE
2  //Nodo rojo mas cercano a un nodo v
3
4  const int N = 2e5+10;
5  vector<int>G[N];
6  int tag[N]; // time of discovery of centroidi
7  int fat[N]; // father in centroid decomposition
8  int szt[N]; // size of subtree
9
10 int best[N];
11 int niv[N];
12 int dist[30][N]; //dist[nivel][nodo];
13
14 void dfs_level(int node, int p, int niv, int prof){
15     dist[niv][node] = prof;
16     for(auto y : G[node]){
17         if(y != p and tag[y] < 0){
18             dfs_level(y, node, niv, prof+1);
19         }
20     }
21 }
22 // -----CENTROID-----
23 int calcsz(int node, int p){
24     szt[node] = 1;
25     for(auto y : G[node]){
26         if(y != p and tag[y] < 0){
27             szt[node] += calcsz(y, node);
28         }
29     }
30     return szt[node];
31 }

```

```

32 int ccnt = 0;
33 void cdfs(int node, int p, int nivel = 0, int sz = -1){
34     if(sz < 0) sz = calcsz(node, -1);
35     for(auto y : G[node]){
36         if(tag[y] < 0 and szt[y] * 2 >= sz){
37             szt[node] = 0;
38             cdfs(y, p, nivel, sz);
39             return;
40         }
41     }
42     tag[node] = ccnt++;
43     fat[node] = p;
44     niv[node] = nivel;
45     dfs_level(node, -1, nivel, 0);
46     for(auto y : G[node]){
47         if(tag[y] < 0){
48             cdfs(y, node, nivel+1);
49         }
50     }
51 }
52
53 int n;
54 void centroid(){
55     ccnt = 0;
56     repl(i,1,n+1){
57         tag[i] = -1;
58     }
59     cdfs(1, -1);
60 }
61
62 //-----
63
64 void update(int node, int rojo){
65     int nivel = niv[node];
66     int cost = dist[nivel][rojo];
67     best[node] = min(best[node], cost);
68     int p = fat[node];
69     if(p > 0){
70         update(p, rojo);
71     }
72 }
73
74 int improve(int node, int cur){
75     if(node <= 0){
76         return 1e9;
77     }
78     int nivel = niv[node];
79     int cost = best[node] + dist[nivel][cur];
80     int p = fat[node];
81     return min(cost, improve(p, cur));
82 }
83
84 void solve(){
85     int qq;
86     cin >> n >> qq;
87     repl(i,0,30){
88         repl(j,1,n+1){
89             dist[i][j] = -1;
90             best[j] = 1e9;
91         }
92     }
93     repl(i,0,n-1){
94         int a, b;
95         cin >> a >> b;
96         G[a].push_back(b);
97         G[b].push_back(a);
98     }
99     centroid();
100    update(1, 1);
101    while(qq--){
102        int q, node;
103        cin >> q >> node;
104        if(q == 1){
105            update(node, node);
106        }
107        else{
108            int res = best[node];
109            int res2 = improve(node, node);
110            cout << min(res, res2) << '\n';
111        }
112    }
113 }
114
115 int main(){
116     velocito;
117     int t = 1;

```

```

118     // cin >> t;
119     while(t--){
120         solve();
121     }
122     return 0;
123 }
124 //Sabrossus
125 //TC Argentina

```

9.4. HLD

```

1  struct HLD {
2      int n, act;
3      vector<vector<int>> adj;
4      vector<int> parent, depth, heavy, head, pos;
5      vector<int> val;
6      vector<int> seg;
7      int tam;
8      HLD(int n) : n(n) {
9          adj.resize(n+1);
10         parent.resize(n+1);
11         depth.resize(n+1);
12         heavy.assign(n+1, -1);
13         head.resize(n+1);
14         pos.resize(n+1);
15         val.resize(n+1);
16         act = 0;
17     }
18
19     void add_edge(int u, int v) {
20         adj[u].push_back(v);
21         adj[v].push_back(u);
22     }
23
24     int dfs(int u, int p) {
25         parent[u]=p;
26         int size=1, max_sub=0;
27         for (int v : adj[u]){
28             if(v==p)continue;
29             depth[v]=depth[u]+1;
30             int sub=dfs(v,u);
31             if(sub>max_sub){
32                 max_sub=sub;
33                 heavy[u]=v;
34             }
35             size+=sub;
36         }
37         return size;
38     }
39
40     void decompose(int u, int h) {
41         head[u]=h;
42         pos[u]=act++;
43         if(heavy[u]!=-1)decompose(heavy[u], h);
44         for(int v: adj[u]){
45             if(v!=parent[u] and v!=heavy[u])decompose(v,v);
46         }
47     }
48
49     void build(){
50         tam=1;
51         while (tam < n) tam<<=1;
52         seg.assign(2*tam,inf);
53         for (int i = 1; i <= n; i++){
54             seg[tam+pos[i]]=val[i];
55         }
56         for (int i = tam - 1; i > 0; i--){
57             seg[i]=max(seg[2*i],seg[2*i+1]);
58         }
59     }
60
61     void update(int u, int value) {
62         int i=tam+pos[u];
63         seg[i]=value;
64         for(i>>=1; i; i>>=1){
65             seg[i]=max(seg[2*i],seg[2*i+1]);
66         }
67     }
68
69     int query(int l, int r) {
70         int res=inf;
71         l+=tam;
72         r+=tam;
73         while(l <= r){

```

```

74         if(l&1)res=max(res,seg[l++]);
75         if(!(r&1))res=max(res,seg[r--]);
76         l>>=1;
77         r>>=1;
78     }
79     return res;
80 }
81
82 void init(){
83     int root=1;
84     dfs(root, 0);
85     decompose(root,root);
86     build();
87 }
88
89 int query_path(int a, int b){
90     int res=inf;
91     while(head[a]!=head[b]){
92         if(depth[head[a]]<depth[head[b]])swap(a,b);
93         res=max(res,query(pos[head[a]],pos[a]));
94         a=parent[head[a]];
95     }
96     if(depth[a]>depth[b])swap(a,b);
97     res=max(res,query(pos[a],pos[b]));
98     return res;
99 }
100 };
101
102 void solve(){
103     int n,q;
104     cin>>n>>q;
105     HLD hld(n);
106     for(int i = 1 ; i <= n ; i++){
107         cin>>hld.val[i];
108     }
109     for(int i = 0; i < n-1 ; i++){
110         int a,b;
111         cin>>a>>b;
112         hld.add_edge(a,b);
113     }
114     hld.init();
115     while(q--){
116         int t;
117         cin>>t;
118         if(t==1){
119             int s,x;
120             cin>>s>>x;
121             hld.update(s,x);
122         }
123         else{
124             int a,b;
125             cin>>a>>b;
126             cout<<hld.query_path(a,b)<<' ' ;
127         }
128     }
129 }

```

9.5. Lowest Common Ancestor

```

1  const int N = 1e5;
2  const int LOG = 20;
3  vector<int> G[N];
4  int depth[N];
5  int up[N][LOG];
6  // Assign levels and get parents
7  void dfs(int u){
8      for(int v : G[u]){
9          depth[v] = depth[u] + 1;
10         up[v][0] = u;
11         for(int j = 1; j < LOG; j++){
12             up[v][j] = up[up[v][j-1]][j-1];
13         }
14         dfs(v);
15     }
16 }
17 int get_lca(int a, int b){
18     if(depth[a] < depth[b]) swap(a,b);
19     int k = depth[a] - depth[b];
20     for(int j = LOG-1; j >= 0; j--){
21         if(k & (1<<j)){
22             a = up[a][j];
23         }
24     }
25     if(a == b) return a;

```



```

29 V 86 W 87 X 88 Y 89 Z 90
30
31 LOWERCASE
32 -----
33 a 97 b 98 c 99 d 100 e 101 f 102 g 103
34 h 104 i 105 j 106 k 107 l 108 m 109 n 110
35 o 111 p 112 q 113 r 114 s 115 t 116 u 117
36 v 118 w 119 x 120 y 121 z 122
37
38 COMMON SYMBOLS
39 -----
40 SPACE 32 ! 33 " 34 ##### # 35 ##### $ 36
41 % 37 ##### & 38 ##### ' 39 ##### ( 40 ##### ) 41
42 * 42 ##### + 43 ##### , 44 ##### - 45 ##### . 46
43 / 47 ##### : 58 ##### ; 59 ##### < 60 ##### = 61
44 > 62 ##### ? 63 ##### @ 64 ##### [ 91 ##### \ 92
45 ] 93 ##### ^ 94 ##### _ 95 ##### ` 96 ##### { 123
46 | 124 ##### } 125 ##### ~ 126
47
48 USEFUL TRICKS
49 -----
50 int ux = uc - '0'; ##### // digit -> integer
51 char c = '0' + ux; ##### // integer -> digit
52
53 int ux = uc - 'a'; ##### // lowercase -> [0,25]
54 int ux = uc - 'A'; ##### // uppercase -> [0,25]
55
56 islower(c)
57 isupper(c)
58 isdigit(c)
59 isalpha(c)
60 isalnum(c)
61 isspace(c)
62 tolower(c)
63 toupper(c)
64 =====

```

10.3. Int128

```

1 // --- LECTURA B SICA ---
2 __int128 read128() {
3     string s;
4     cin >> s;
5     __int128 n = 0;
6     int sign = 1;
7     size_t start = 0;
8
9     if (!s.empty() && s[0] == '-') {
10         sign = -1;
11         start = 1;
12     }
13
14     for (size_t i = start; i < s.length(); ++i) {
15         n = n * 10 + (s[i] - '0');
16     }
17
18     return n * sign;
19 }
20
21 // --- IMPRESI N B SICA ---
22 void print128(__int128 n) {
23     if (n == 0) {
24         cout << 0;
25         return;
26     }
27     if (n < 0) {
28         cout << '-';
29         n = -n;
30     }
31     string s;
32     while (n > 0) {
33         s += (char)('0' + (n % 10));
34         n /= 10;
35     }
36     reverse(s.begin(), s.end());
37     cout << s;
38 }

```

10.4. Merge Sort

```

1 // Rango de uso: 0-indexed.
2 // Llamar en el solve como: mergeSort(v, 0, n - 1);
3 ll ans = 0;
4
5 void mergeSort(vector<ll> &v, int low, int high) {
6     if (low == high) return;
7
8     int mid = (low + high) >> 1;
9     mergeSort(v, low, mid);
10    mergeSort(v, mid + 1, high);
11
12    queue<ll> L, H;
13    for (int i = low; i < mid + 1; i++) {
14        L.push(v[i]);
15    }
16    for (int i = mid + 1; i < high + 1; i++) {
17        H.push(v[i]);
18    }
19
20    for (int i = low; i < high + 1; i++) {
21        if (L.size() == 0) {
22            v[i] = H.front();
23            H.pop();
24        }
25        else if (H.size() == 0) {
26            v[i] = L.front();
27            L.pop();
28        }
29        else {
30            if (L.front() <= H.front()) {
31                v[i] = L.front();
32                L.pop();
33            }
34            else {
35                v[i] = H.front();
36                H.pop();
37                ans += L.size(); // Inversion detectada
38            }
39        }
40    }
41 }

```

10.5. Prefix Sum 2D

```

1 //1 indexed
2 const int N = 1005;
3 int v[N][N];
4 int acu[N][N];
5 int n, m;
6
7 void solve() {
8     //limpiar la acu para cada TC
9
10    // Construcción: acumula 1 si el elemento es exactamente 1
11    for (int i = 1; i <= n; i++) {
12        for (int j = 1; j <= m; j++) {
13            int val = (v[i][j] == 1 ? 1 : 0);
14            acu[i][j] = val + acu[i - 1][j] + acu[i][j - 1] - acu[i - 1][j - 1];
15        }
16    }
17
18    // Esquinas de consulta:
19    // (a, b) -> Superior Izquierda
20    // (c, d) -> Inferior Derecha
21    int a = 2, b = 2, c = 4, d = 4; // Valores de ejemplo
22
23    int ans = acu[c][d] - acu[a - 1][d] - acu[c][b - 1] + acu[a - 1][b - 1];
24    cout << ans << '\n';
25 }

```

10.6. Priority Queue Min Heap

```

1 priority_queue<int, vector<int>, greater<int>> pq;

```

10.7. Random

```

1 mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
2 // mt19937_64 para 64 bits usar uint64_t o unsigned long long
3 cout << rng() << endl;
4 shuffle(v.begin(),v.end(),rng);

```

10.8. Plantilla Python

```

1 import sys
2 from bisect import bisect_left, bisect_right, insort
3 from collections import defaultdict, Counter, deque
4 from heapq import heappush, heappop, heapify
5 from math import gcd, lcm, isqrt
6
7 input = sys.stdin.readline
8
9
10 def solve():
11     n = int(input()) #leer numero
12     a,b,c= map(int, input().split()) #leer numeros
13     s = input().strip() #leer cadenas
14     v = list(map(int, input().split())) #leer lista de enteros
15
16
17 t = 1
18 # t = int(input())
19 for _ in range(t):
20     solve()
21
22 # C++                                Python
23
24 # vector                             list
25 # array                             list
26 # string                             str
27
28 # set                                set
29 # map                                dict
30 # multiset                           Counter
31 # unordered_map                      dict
32 # unordered_set                     set
33
34 # priority_queue                     heapq
35 # queue                              deque
36 # deque                              deque
37
38 # lower_bound                        bisect_left
39 # upper_bound                        bisect_right
40 # sort                               list.sort()
41 # binary_search                      bisect_left/right
42
43 # pair                               tuple
44 # struct                             class / dataclass
45
46 # gcd                                math.gcd
47 # lcm                                math.lcm
48
49 # push_back                          append
50 # push_front                         appendleft
51 # pop_back                           pop
52 # pop_front                         popleft
53
54 # size()                             len()
55 # empty()                            not x

```

10.9. Pragma

```

1 #pragma GCC optimize("O3,unroll-loops")
2 #pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")

```